

**Film Investment Screener: A Multimodal Machine-Learning Tool for Box-Office
Prediction and Investment Decision Support**

Zhiheng Zhou

The University of Chicago MAPSS

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Abstract

This capstone project develops a machine-learning decision-support tool for film investment by translating box-office predictions into investor-oriented screening metrics. It tests whether multimodal film data, including plot text, genre, production details, director track records, release timing, and early audience signals, can help identify films with stronger commercial potential. Research indicates that combining features related to “who,” “what,” and “when” yields better predictions (Lash & Zhao, 2016). The project includes a supervised learning pipeline to predict logarithmic box-office revenue and a profitability classifier to assess whether a film will earn over 2.5 times its budget. An interactive Gradio web interface, deployed on Hugging Face Spaces, allows users to input a film concept and receive predicted revenue ranges, profitability probabilities, comparable films, and risk flags. While the model proves useful as an investment screener, accurately predicting revenue remains challenging for certain film categories.

Introduction

Investing in films is inherently uncertain because a film's commercial performance relies on a mix of creative, industrial, and market factors. Investors are generally not focused on predicting the exact box-office numbers. Instead, they need to understand whether a project is likely to meet certain financial thresholds, if it falls within a commercially viable genre or release window, and how its risk profile compares to those of past successful or unsuccessful films. My capstone project addresses this challenge by developing a machine-learning system that translates film attributes into predictions relevant to investors. For me, the central problem was not simply whether a model could predict box-office revenue, but whether the prediction could be translated into a decision that a potential investor could use.

The research question guiding this project is: Can a multimodal machine-learning model that incorporates a film's plot, genre, cast, and director metadata, along with early audience signals, accurately predict box-office performance to create actionable investment decision rules for potential film investors? This question positions the project within media economics and frames box-office forecasting as both a supervised prediction problem and an investment decision problem.

Previous research highlights the challenges and potential benefits of predicting box-office performance. De Vany and Walls (1999) found that movie revenues are highly uncertain and follow a heavy-tailed distribution, where rare blockbuster successes greatly influence overall revenue. This finding illustrates the difficulty of making precise revenue predictions and suggests that investment-focused models

should prioritize not only point estimates but also consider risk, ranking, and probabilities. Similarly, Sharda and Delen (2006) approach box-office prediction as a decision-support problem by transforming revenue forecasting into a classification task. They demonstrate that machine learning models can effectively categorize films into various performance ranges before their release. Additionally, Eliashberg et al. (2007) extend this decision-support approach to the green-lighting process by using textual information from storylines to forecast return on investment (ROI) and support production decisions.

More recent work suggests one way to manage this uncertainty: combining multiple kinds of signals rather than relying on a single feature group. Lash and Zhao (2016) argue that predicting early movie investments is more effective when combining various feature groups, including the individuals involved in the film, the film's content, and the timing of its release. Qiu and Zheng (2023) demonstrate that integrating sentiment-driven forecasts can enhance box-office projections, especially when model uncertainty is carefully addressed. Similarly, Cheng and Yang (2022) find that both online review sentiment and review volume can improve economic models of box-office sales. Collectively, these studies suggest that predicting film performance should not rely solely on genre or budget; instead, it should incorporate textual analysis, structured metadata, audience feedback, and financial variables.

Data and Feature Construction

The final dataset was constructed using three primary sources: Box Office Mojo

yearly CSV files, MovieLens 25M metadata, and the TMDB API. The Box Office Mojo files provided information about domestic, foreign, and worldwide revenue for approximately 5,000 film records. This included 2,000 entries from the 2000s (2000-2009), 2,800 from the 2010s and early 2020s (2010-2023), and 200 from 2024. After cleaning the titles and removing duplicates, the dataset retained 4,999 unique films. MovieLens served as an initial source of genre information via title matching, while the TMDB API provided key enrichment fields. These fields included budget, runtime, release date, overview, tagline, director, top cast, production details, keywords, popularity, vote average, and vote count. Because profitability classification depends on production budget, analyses involving ROI or the $2.5\times$ budget threshold were limited to films with valid budget information.

The final modeling dataset merged textual, categorical, and numerical features associated with each film. Textual features were generated by combining genres, keywords, overviews, and taglines into a single text field, which was then vectorized using TF-IDF. Numerical features included the logarithm of the budget, runtime, logarithm of the vote count, vote average, popularity, average previous revenue of directors, the count of previous films directed, release month, and binary indicators for summer or holiday release windows. Categorical features encompassed the primary genre, original language, and release quarter. The resulting feature matrix integrated textual, numerical, and categorical information into a cohesive format suitable for modeling.

This multimodal approach addresses a common limitation found in simpler text-

only prediction models. While models that rely primarily on genres and keywords can capture some aspects of film content, they often overlook crucial financial factors such as budget, runtime, release timing, director history, and audience reception. Therefore, this model extends the feature set to include structured and categorical variables. This adjustment aligns the model more closely with actual investment considerations, as revenue alone becomes less meaningful without also accounting for budget and expected returns.

Methods

I approached the project as two interconnected supervised learning tasks. The first task was a regression model predicting log-transformed box-office revenue. A log transformation was necessary because film revenue is highly skewed, with a few blockbuster films earning significantly more than most other releases. This approach aligns with prior research that indicates box-office outcomes are heavy-tailed and often dominated by rare blockbuster successes (De Vany & Walls, 1999). The second task was classification, where I aimed to predict whether a film is likely to be profitable, defined as earning more than 2.5 times its production budget. This profitability threshold serves as a simplified rule of thumb, making the output easier to interpret for investment screening. Framing the task partly as classification follows earlier box-office prediction research that treated film performance categories as more useful decision-support outputs than exact revenue estimates (Sharda & Delen, 2006).

In the modeling pipeline, I compared text-only baselines with fused multimodal

models. The text-only baseline used TF-IDF features with a linear model. In contrast, the fused models combined TF-IDF text features with scaled numerical features and one-hot-encoded categorical features. The strongest model employed a gradient boosting regressor on the fused feature set. Additionally, I trained a logistic regression classifier to determine profitability for films with known budgets. Beyond standard regression metrics, such as R^2 and error measures, I also evaluated the model using investor-oriented metrics, including top-decile precision, decile lift, and simulated return on investment.

A key methodological choice in this project is the implementation of time-aware evaluation. Instead of using a random train/test split, which could overstate performance by combining films from very different market conditions on both sides, I opted for a chronological split. This means that I trained the model on earlier films and evaluated it on more recent releases. This approach more accurately simulates the real-world scenario in which investors must decide on future releases solely based on past data. This choice is especially important because the theatrical market has changed substantially in recent years, including the rise of streaming releases, pandemic disruptions, and shifts in distribution strategy.

Results

The results indicate that the model is more effective as a screening and ranking tool than as an accurate revenue forecaster. In time-aware testing, the text-only TF-IDF baseline underperformed, resulting in a negative test R^2 value. This suggests that while

text-only models may show strong performance on training data, this performance can be misleading due to temporal leakage or similarities between randomly mixed films. Under a realistic chronological split, relying solely on text features is inadequate for generalizing to newer releases. This result became one of the main reasons I shifted the project away from exact revenue prediction and toward screening, ranking, and risk interpretation.

In contrast, the fused multimodal model performed substantially better. The best-performing model was the gradient boosting model that utilized a combination of text, numerical, and categorical features. In the final notebook summary, this model achieved a test R^2 of approximately 0.33 on log revenue, with a precision of 73% among the top 10% of predicted films, a decile lift of about $5.9\times$, and a simulated ROI of $3.57\times$ on a top 25 portfolio. These results suggest that, even though the model may not perfectly predict exact revenue, it can effectively rank films for portfolio-style investment screening.

The profitability classifier performed particularly well, achieving a validation AUC of approximately 0.80 and a test AUC of around 0.88. Since this classifier predicts whether a film surpasses the $2.5\times$ budget threshold, its output is more actionable than raw revenue predictions. A film with high predicted profitability can be considered a stronger candidate for further development, while a film with low probability may be flagged for rejection or a more in-depth qualitative review.

Finally, the unsupervised analysis revealed interpretable genre-level patterns. K-means clustering and LDA topic modeling indicated that family animation, fantasy

adventure, action, and science-fiction content tend to occupy more commercially promising areas of the dataset. In contrast, romance and smaller-scale drama clusters show lower average revenue. These findings do not imply that genre causes revenue, but they do demonstrate that content patterns derived from text features correspond to meaningful commercial differences.

Interactive User Interface

A key deliverable of this capstone project is the Film Investment Screener, which features an interactive Gradio interface. This user-friendly UI enables users to input various details about a hypothetical film, including the title, keywords, budget, release month, primary genre, logline, tagline, runtime, original language, director's track record, prior films, vote count, vote average, and popularity. Once the user submits the information, the app generates four outputs: predicted revenue along with an 80% confidence interval, the probability of profitability, comparable historical films, and risk flags. The interface utilizes a trained gradient boosting model, a profitability classifier, feature pipelines, residuals, and a reference film dataset sourced from saved artifacts.

The application is designed to be accessible for non-technical users. Rather than presenting only numerical predictions, it offers a comprehensive screening report featuring visual cards. This includes a green/yellow/red profitability interpretation, a break-even target, and comparable films identified through TF-IDF cosine similarity. The risk flags are based on known model failure modes, such as missing budget

information, extremely low or high budgets, non-English films, less favorable release months, high-budget drama or romance projects, unknown directors, and unusually wide confidence intervals. The final user interface was uploaded to both GitHub and Hugging Face Spaces and can be accessed at <https://huggingface.co/spaces/joezhou21/film-investment-screener>. Figure 1 illustrates the deployed application using a hypothetical animated film, Sky Pirates, as a worked example. Panel (a) shows the required input fields that drive the prediction: title, keywords, budget, release month, and primary genre. Panel (b) shows the optional inputs that enrich the model’s text and metadata signals, including logline, tagline, runtime, original language, the director’s track record, and pre-release audience buzz. Panel (c) shows the screening report returned after the user clicks Run Screening: a predicted revenue of \$155M with an 80% confidence interval of \$91.3M–\$254.0M, a 94% profitability probability with a green “Greenlight” interpretation against a \$200M break-even target, an automated risk assessment, and the top five historically comparable films ranked by text similarity.

The screenshot displays the 'Film Investment Screener' web application interface. At the top, it shows the Hugging Face Spaces header with the app name 'film-investment-screener' and a 'Run' button. Below the header, the app title 'Film Investment Screener' is followed by a brief description: 'Multimodal box-office forecasting. Enter a hypothetical film concept; get a revenue estimate, profitability probability, comparable historical films, and risk flags.' A 'Clear all fields' button is located below the description. A section titled 'REQUIRED — THESE DRIVE THE PREDICTION' contains five input fields, each with a close button (x) on the right:

- Title:** A text input field containing 'Sky Pirates'.
- Keywords (English, space-separated):** A text input field containing 'family animation adventure fantasy magic children friendship pirate sky'. A small note below the field reads: 'The single most important input. These drive the model's understanding of genre/theme.'
- Budget (\$M):** A numeric input field containing '80'. A small note below the field reads: 'Production budget in millions USD.'
- Release Month:** A dropdown menu showing 'Nov' with a green upward arrow icon.
- Primary Genre:** A dropdown menu showing 'animation'.

Figure 1a. Required input fields.

Story details — used as additional text features.

Logline / Synopsis
A young girl discovers a hidden city in the clouds and joins a band of sky pirates.

Tagline
Adventure is in the air.

Production details.

Runtime (min)
95

Original language
en

Director track record. Use defaults if unknown.

Director's avg log-revenue (prior films)
14 = flop director · 18.5 = median · 21+ = blockbuster director
18.5

Number of prior films directed
3

Pre-release buzz (leave defaults if unknown).

Pre-release vote count (TMDB-style audience engagement)
2500

Expected quality rating (1-10)
7.5

TMDB popularity
35

Figure 1b. Optional input fields.

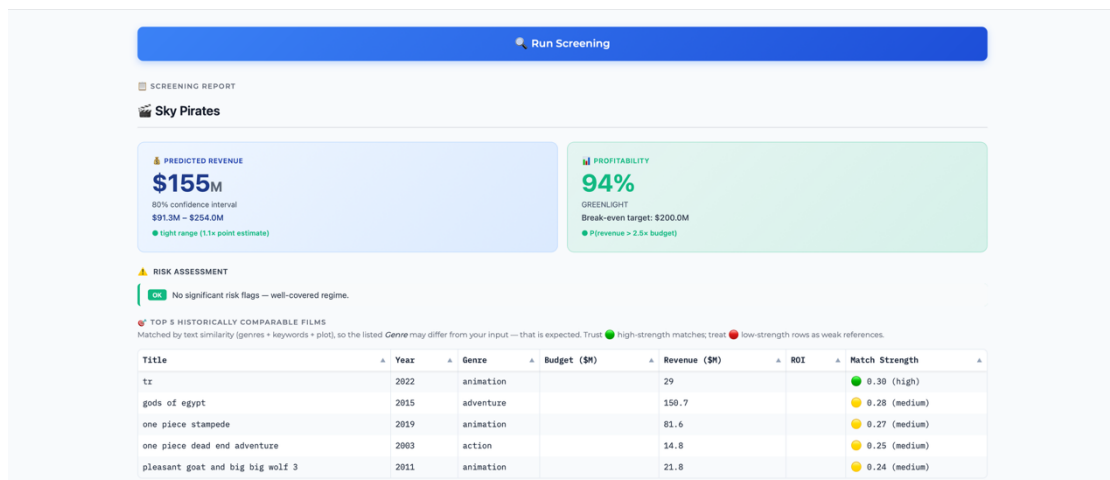


Figure 1c. Screening report output.

Figure 1. The Film Investment Screener deployed on Hugging Face Spaces. (a) Required inputs. (b) Optional story, production, director, and pre-release buzz inputs. (c) Screening report for the example film Sky Pirates, including predicted revenue with confidence interval, profitability probability, risk flags, and comparable historical films.

Discussion

The main finding of this capstone project is that multimodal machine learning can support film investment decisions, but only when the task is framed realistically. In this sense, the project shifted focus from finding a perfect forecasting model to developing a structured approach for distinguishing more promising projects from riskier ones. Predicting box-office performance with precision remains challenging because of the numerous factors that influence film success, many of which are not fully captured in public datasets. These factors include marketing expenditure, distribution strategy, star publicity, franchise expectations, platform competition, and cultural timing. However, investors do not necessarily require a perfect point prediction. Instead, they need a model that can rank projects, flag risks, identify comparable films, and estimate whether a film has a reasonable chance of becoming profitable.

This project argues that incorporating multimodal features is essential for meaningful predictions. Text features help the model understand the film's content, while budget and release metadata provide critical insights into scale, distribution goals, and market positioning. Additionally, information about the director's history and audience signals helps better understand the expected commercial potential. This aligns with Lash and Zhao's (2016) focus on the importance of "who," "what," and "when" features, as well as the value of sentiment and review-based variables highlighted in more recent research (Cheng & Yang, 2022; Qiu & Zheng, 2023).

Furthermore, this project highlights the importance of time-aware evaluation. A random split of data can create an illusion of model strength, as films from similar periods and market conditions may appear in both the training and test sets. The film

market post-2020 is markedly different due to the rise of streaming releases, pandemic disruptions, changes in theatrical attendance, and varying distribution strategies. By utilizing a chronological split, this capstone provides a more accurate estimate of how the model would perform in a real investment scenario.

Implications

This project has practical implications for the use of machine learning in film development and investment decisions. The Film Investment Screener serves as an early-stage tool that helps compare hypothetical film concepts, identify commercially promising patterns, and flag projects that may require closer examination. For example, it can help users evaluate whether a film's genre, budget, release timing, and textual profile resemble those of past high-performing films. It also provides warnings when predictions may be less reliable due to missing budget data, non-English-language films, or unusually wide confidence intervals.

Additionally, the project suggests that film investment should be assessed from both predictive and interpretive perspectives. Even if a model cannot perfectly predict exact revenue, it can still help formulate better investment questions, such as: What comparable films does this project resemble? Is the budget appropriate for the genre? Does the release window increase or decrease risk? Is the predicted profitability strong enough to justify further development? In this way, the screener's value lies not only in providing forecasts but also in organizing the decision-making process around comparisons of risk and expected return.

More broadly, this capstone demonstrates how academic prediction models can be transformed into accessible tools for creative industries. By deploying the model through a Gradio interface on Hugging Face Spaces, the project converts a technical notebook into a user-friendly platform that non-technical users can test and interpret. This approach is significant because cultural products are influenced by timing, taste, marketing, and human judgment in ways that public data alone cannot fully capture. Thus, machine learning may be most valuable in creative markets when it supports exploration and risk awareness rather than asserting final decisions.

Limitations

Several limitations are present in the model. First, it does not account for marketing spend, which is one of the most significant predictors of box-office performance, yet such data is rarely available in public datasets. Second, budget information is often incomplete, particularly for independent and international films. Since profitability classification relies on the budget, the classifier performs best for films with more comprehensive production data. Third, the model may overpredict the success of re-releases or films with substantial historical popularity, as the popularity metrics may reflect the original release rather than the specific theatrical run being evaluated. Fourth, the model's reliability decreases for non-English films and limited-distribution releases, where global popularity metrics may not directly correlate with U.S. or worldwide theatrical revenue. Lastly, the model is correlational rather than causal. A high-budget film might generate more revenue not solely

because of its budget, but also because it is linked to studio support, marketing, distribution, talent, and franchise value. Consequently, the model should be utilized as a decision-support tool rather than as a definitive greenlight system.

References

- Cheng, L.-C., & Yang, Y. (2022). The effect of online reviews on movie box office sales: An integration of aspect-based sentiment analysis and economic modeling. *Journal of Global Information Management*, 30(1), 1–16. <https://doi.org/10.4018/JGIM.298652>
- De Vany, A., & Walls, W. D. (1999). Uncertainty in the movie industry: Does star power reduce the terror of the box office? *Journal of Cultural Economics*, 23(4), 285–318. <https://doi.org/10.1023/A:1007608125988>
- Eliashberg, J., Hui, S. K., & Zhang, Z. J. (2007). From story line to box office: A new approach for green-lighting movie scripts. *Management Science*, 53(6), 881–893. <https://doi.org/10.1287/mnsc.1060.0668>
- Lash, M. T., & Zhao, K. (2016). Early predictions of movie success: The who, what, and when of profitability. *Journal of Management Information Systems*, 33(3), 874–903. <https://doi.org/10.1080/07421222.2016.1243969>
- Qiu, Y., & Zheng, Y. (2023). Improving box office projections through sentiment analysis: Insights from regularization-based forecast combinations. *Economic Modelling*, 125, 106349. <https://doi.org/10.1016/j.econmod.2023.106349>
- Sharda, R., & Delen, D. (2006). Predicting box-office success of motion pictures with neural networks. *Expert Systems with Applications*, 30(2), 243–254. <https://doi.org/10.1016/j.eswa.2005.07.018>